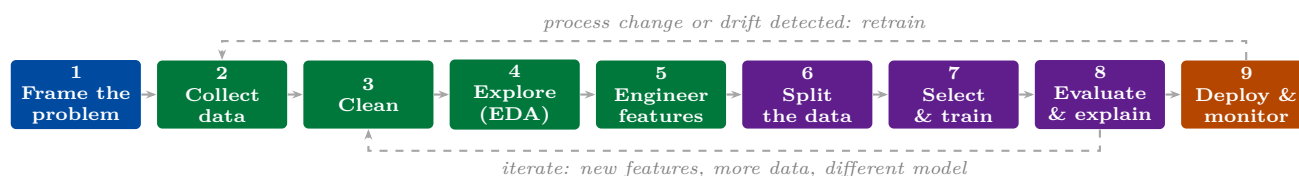


Machine Learning Model Development Guideline

From Plant Problem to Monitored Model: Steps, Checks and Common Pitfalls

Industrial AI Capability Building Workshop · Companion to *Classical Machine Learning Model Selection*



Stage	Key Actions	Common Pitfalls
1 Frame the problem	Name the decision the model will support and who will act on it. Define the target, the prediction horizon and a success criterion in plant terms (e.g. “flag 80% of failing batches 2 h ahead”). Record the current baseline (operator rule, lab check).	Predicting something nobody can act on; using inputs that are not available at the time the prediction is needed.
2 Collect data	Pull data from the historian/SCADA, LIMS, MES and maintenance logs. Align timestamps and sampling rates; allow for residence times and lab-result delays. Walk the line and talk to operators about sensors, recipes and known events.	Matching a lab result to the wrong process time; ignoring recalibrations, sensor swaps and recipe changes.
3 Clean	Convert units; remove duplicates; flag frozen (flat-lined) and out-of-range sensors. Decide how to treat missing values based on <i>why</i> they are missing. Exclude start-ups, shutdowns and CIP if out of scope. Write every rule down.	Deleting outliers automatically: some are sensor faults, others are the real process events you want to predict.
4 Explore (EDA)	Plot every variable over time and as distributions; check target vs. inputs, correlations, class balance, and differences by product, line and shift. Sanity-check patterns with process experts.	Reading correlation as cause; missing hidden drivers such as product grade or season.
5 Engineer features	Build physically meaningful inputs: deviation from setpoint, rates of change, ratios, lagged and rolling values, time since CIP or maintenance. Encode categories (product, line).	Features that use future data (e.g. centred rolling averages); too many near-duplicate features.
6 Split the data	Split by time: train on earlier data, validate and test on later data. Keep whole batches or campaigns in one set. Lock the test set away until the end. Fit imputation and scaling on the training set only.	Random splits of time-series data, which leak future information and overstate accuracy.
7 Select & train	Start with a simple baseline (current rule or linear model), then try tree ensembles (see the <i>Model Selection</i> guide). Tune settings with time-ordered cross-validation on the training data. Prefer the simpler model if the gain is small.	Tuning against the test set; choosing a complex model for a gain within measurement noise.
8 Evaluate & explain	Test once on the held-out set. Report errors in engineering units (MAE, RMSE) or precision/recall for faults, always against the baseline. Break results down by product and operating regime. Check SHAP directions against process knowledge; confirm with a plant trial.	One average score hiding poor performance in the regimes that matter; accepting explanations that contradict the physics.
9 Deploy & monitor	Use the same data pipeline as in training. State the valid operating window. Monitor input drift and live accuracy; set retraining triggers, an owner and a fallback. Version data, code and model; keep a short model record.	Silent degradation after a process change (new equipment, recipe or sensor); no one owning the model after hand-over.

Before Go-Live: Sign-Off Checklist

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|---|---|
| ☐ Beats the current baseline on the untouched test set | ☐ Valid operating window and known limits documented |
| ☐ Performs acceptably for every key product and regime | ☐ Cleaning rules, data sources and versions recorded |
| ☐ Every input is available in real time at prediction | ☐ Drift monitoring, retraining trigger and fallback in place |
| ☐ Explanations agree with process engineers’ understanding | ☐ Named owner accepted by operations |